

```
import pandas as pd
import matplotlib.pyplot as plt

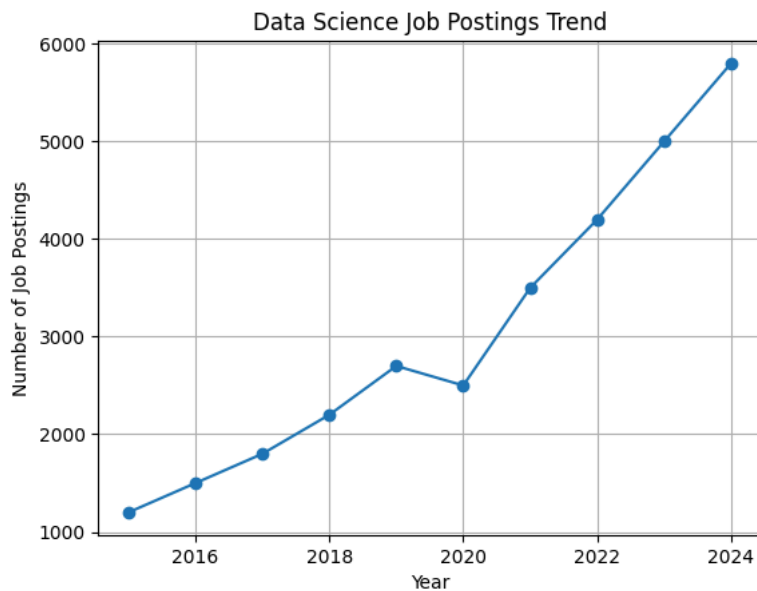
data = {
    'Year': [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024],
    'Job_Postings': [1200, 1500, 1800, 2200, 2700, 2500, 3500, 4200, 5000, 5800]
}

df = pd.DataFrame(data)

print(df)

plt.plot(df['Year'], df['Job_Postings'], marker='o')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.title('Data Science Job Postings Trend')
plt.grid(True)
plt.show()
```

	Year	Job_Postings
0	2015	1200
1	2016	1500
2	2017	1800
3	2018	2200
4	2019	2700
5	2020	2500
6	2021	3500
7	2022	4200
8	2023	5000
9	2024	5800



```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'Role': ['Data Analyst', 'Data Scientist', 'Data Engineer', 'Data Analyst', 'Data Scientist']
})

df['Role'].value_counts().plot(kind='bar')
plt.xlabel('Role')
plt.ylabel('Count')
plt.title('Data Science Roles')
plt.show()
```



```
import pandas as pd

structured = pd.DataFrame({
    'Name': ['A', 'B', 'C'],
    'Age': [20, 21, 19]
})

semi = {'Name': 'A', 'Skills': ['Python', 'SQL']}

unstructured = "Data Science is the study of data."

print("Structured:\n", structured)
print("\nSemi-structured:\n", semi)
print("\nUnstructured:\n", unstructured)
```

```
Structured:
   Name  Age
0     A   20
1     B   21
2     C   19
```

```
Semi-structured:
{'Name': 'A', 'Skills': ['Python', 'SQL']}
```

```
Unstructured:
Data Science is the study of data.
```

```
from cryptography.fernet import Fernet
```

```
key = Fernet.generate_key()
f = Fernet(key)
```

```
data = "Sensitive Data"
encrypted = f.encrypt(data.encode())
decrypted = f.decrypt(encrypted).decode()
```

```
print("Encrypted:", encrypted)
print("Decrypted:", decrypted)
```

```
Encrypted: b'gAAAAABqqg10Vjmp-myYzIwcXAf9ihBot74D-t-I0c_DHm99Tz6H1oVQQvyT_5iT1jetDSJLEc3lQ2oYhcKVn1Nj3iB8-83MDQ=='
Decrypted: Sensitive Data
```